Exercise 14

On relative positions of two planes in space

From the school book | Understand | Apply | Higher Order Thinking Skills

1 **Find the cosine of the angle between each pair of the following planes :**

(1) $4x + 4y - 2z = 9$, $2x + y + z = -3$ « $\frac{5\sqrt{6}}{18}$ »

(2) $\vec{r} \cdot (2, -1, 4) = 5$, $3x - y + 2z = 4$ « $\frac{5\sqrt{6}}{14}$ »

(3) $\vec{r} \cdot (2, -1, 2) = 3$, $\vec{r} \cdot (3, 2, -6) = 7$ « $\frac{8}{21}$ »

2 **Find the measure of the angle between each pair of the following planes :**

(1) $\vec{r} \cdot (2, 1, -1) = 4$, $\vec{r} \cdot (3, -2, 0) = 7$ « 63° 4′ 10″ »

(2) $y = 4$, $x - 3y + 5z = 1$ « 59° 31′ 47″ »

(3) $3x - y = 5$, $x - 2y = 4$ « 45° »

(4) $x - 3y + 2z = 0$, $2x + y - z = 3$ « 70° 35′ 36″ »

3 **Determine the value of each of k and ℓ If each pair of the following planes is parallel :**

(1) $2x - y + kz = 5$, $x + \ell y + 4z = 1$ « $8, -\frac{1}{2}$ »

(2) $3x - ky + 5z - 4 = 0$, $6x + \ell z = 7$ « 0 , 10 »

(3) $\vec{r} \cdot (-2, k, 5\ell) = -2$, $5x + 4y + 20z = 8$ « – 1.6 , – 1.6 »

4 **Determine the value of a that makes each pair of the following planes at it is perpendicular :**

(1) $3x - 4y + 2z + 9 = 0$, $3x + 4y - az + 7 = 0$ « – 3.5 »

(2) $x - 3y + z = 4$, $ax + 2y + 3z = 2$ « 3 »

(3) $(3, -5, a) \cdot \vec{r} = 3$, $x + 3y + 2z = -5$ « 6 »

(4) $7x - 2y - 7 = 0$, $(a, 1, -3) \cdot \vec{r} = 0$ « $\frac{2}{7}$ »

5 Determine the position of each of the two planes with respect to the other :

$2x + 3y + 4z - 12 = 0$, $3x - 6y + 1 = 0$

6 **Choose the correct answer from those given :**

(1) The measure of the angle included between the two planes :

$x + y - 1 = 0$, $y + z - 1 = 0$ equals .....

(a) 30° (b) 45° (c) 60° (d) 75°

(2) The measure of the angle between the two planes

$3x - 4y + 5z = 0$, $2x - y - 2z = 5$ is .....

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

(3) *(1st Session 2022)* The measure of the angle between the two planes $x + y + 2z = 7$, $\vec{r} \cdot (2, -1, 1) = 6$ equals .....

(a) 30° (b) 45° (c) 60° (d) 90°

(4) *(2nd Session 2019)* The measure of the angle between the two planes :

$3x - 6y + 6z - 4 = 0$, $x + z - 7 = 0$ is .....

(a) 90° (b) 60° (c) 45° (d) 30°

(5) The measure of the angle between the two planes $y = 0$ and $z = 0$ is .....

(a) 30° (b) 45° (c) 60° (d) 90°

(6) The measure of the angle between the two planes $x = 5$ and $y = -4$ is .....

(a) 30° (b) 60° (c) 90° (d) 120°

(7) *(2nd Session 2021)* The measure of the angle between the plane xy and the plane $x + \sqrt{3}z - 7 = 0$ equals .....

(a) 60° (b) 90° (c) 30° (d) 45°

(8) The two planes $-2x + 3y - 5z = 3$, $-4x + 6y - 10z = 10$ are .....

(a) intersect. (b) parallel. (c) coincide. (d) perpendicular.

(9) The two planes $2x - y + 2z = 8$, $3x + 4y - z - 7 = 0$ are .....

(a) perpendicular. (b) parallel.

(c) coincide. (d) the measure of their included angle = $\frac{\pi}{4}$

(10) *(1st Session 2019)* If the two planes : $3x - y + 2z + 4 = 0$, $x + 2y + kz = 2$ are perpendicular , then k = ...

(a) -4 (b) $\frac{2}{3}$ (c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

(11) *(1st Session 2022)* If the two planes : $\vec{r} \cdot (1, m, 16) = 5$ and $\ell x + 2y + 8z = 8$ are parallel , then ℓm = ...

(a) 8 (b) 2 (c) $\frac{1}{2}$ (d) $\frac{1}{8}$

(12) *(1st Session 2021)* If the two planes $2x + cy + 4z = 1$ and $(a + 2)x + 6y + (b - 2)z = 5$ are parallel , then $2a - b$ = ...

(a) -6 (b) 6 (c) -12 (d) 12

(13) The two planes : $2x + 3y + 4z - 24 = 0$, $\vec{r} \cdot (2, -4, 0) = -1$ are .....

(a) coincident. (b) parallel.

(c) intersecting and perpendicular. (d) intersecting but not perpendicular.

(14) If the plane $ax + by + 3z = 1$ is perpendicular to both the two planes $3x - 4y + 2z = 0$ and $2x - 3y - 5 = 0$, then $a - b =$ .....

(a) -21 (b) -6 (c) 6 (d) 21

(15) The distance between the point $(3, -1, 2)$ and the xz-plane equals ..... length units.

(a) 3 (b) -1 (c) 2 (d) 1

(16) The two planes $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are perpendicular if .....

(a) $a_1a_2 + b_1b_2 + c_1c_2 = 1$ (b) $a_1a_2 + b_1b_2 + c_1c_2 = 0$

(c) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ (d) $a_1a_2 + b_1b_2 + c_1c_2 + d_1d_2 = 0$

(17) The two planes : $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$ are parallel if .....

(a) $a_1a_2 + b_1b_2 + c_1c_2 = 0$ (b) $a_1 = m\,a_2$, $b_1 = mb_2$, $c_1 = m\,c_2$

(c) $a_1 = a_2$, $b_1 = b_2$ (d) $\frac{a_1}{a_2} = \frac{b_2}{b_1} = \frac{c_1}{c_2}$

(18) The equation of the line of intersection of the two planes P_1 : $2x - y + z - 1 = 0$, $P_2 : x - 3y - z + 2 = 0$ is .....

(a) $\frac{x+1}{-1} = \frac{y}{2} = \frac{z}{3}$ (b) $\frac{x-1}{1} = -\frac{y}{3} = \frac{z-5}{1}$

(c) $\frac{x-2}{1} = \frac{y-3}{-2} = \frac{z}{-1}$ (d) $\frac{x-1}{4} = \frac{y-1}{3} = \frac{z}{-5}$

(19) The line of intersection of the two planes $y = 5$, $x = 4$ .....

(a) parallel to the plane $z = 0$ (b) has direction vector $(4, 5, 0)$

(c) has direction vector $(0, 0, 9)$ (d) lies in the xy-plane.

(20) The line of intersection of the two planes $\vec{r} \cdot (3, -1, 1) = 1$, $\vec{r} \cdot (1, 1, -2) = 2$ is parallel to the vector .....

(a) $(2, 7, 1)$ (b) $(-2, 7, 13)$ (c) $(1, 7, 4)$ (d) $(7, 13, 1)$

(21) *(1st Session 2018)* The length of the perpendicular from point $(2, 3, 1)$ to the plane $2x - 2y + z = 5$ is .....

(a) 1 (b) 2 (c) 3 (d) 4

(22) The distance between the plane $2x - 3y + 6z + 14 = 0$ and the origin point is ..... length unit.

(a) 11 (b) 2 (c) – 2 (d) 14

(23) The distance between the two planes $y = 4$, $y = -2$ is ..... length unit.

(a) 3 (b) 2 (c) 6 (d) 8

(24) If the distance between the two planes $y = 4$ and $y = a$ equals 6 length unit , then a = ...

(a) 10 (b) – 2 (c) 10 or – 2 (d) 2

(25) The distance between the two planes : $x - 2y - 2z - 12 = 0$, $\vec{r} \cdot (-3, 6, 6) = 9$ equals ..... length unit.

(a) 2 (b) 3 (c) 4 (d) 5

(26) The length of the perpendicular drawn between the two planes $3x + 12y - 4z = 9$, $3x + 12y - 4z = -17$ equals .....

(a) 2 (b) 3 (c) 4 (d) 5

(27) *(2nd Session 2024)* If the length of the perpendicular drawn from the point $(2, 0, a)$ to the plane $\vec{n} \cdot (1, 2, -1) = 2$ equals $\sqrt{6}$ length unit , then $a = \pm$ .....

(a) 6 (b) $\sqrt{6}$ (c) $\sqrt{2}$ (d) 1

(28) If the length of the perpendicular drawn from the origin point to the plane P is 7 length units and the direction ratios of the straight line that carries it are $-3, 2, 6$ which of the following equations is the equation of the plane P ?

(a) $-3x + 2y + 6z - 7 = 0$ (b) $-3x + 2y + 6z - 49 = 0$

(c) $3x - 2y + 6z + 7 = 0$ (d) $-3x + 2y - 6z - 49 = 0$

(29) The system of equations $x - 2y + z = 5$, $2x + y - 3z = 1$ represents .....

(a) a plane. (b) a circle. (c) a straight line. (d) a point.

(30) The plane y z divides $\overline{AB}$ where A $(2, 3, 1)$, B $(-7, 6, 1)$ by ratio .....

(a) $\frac{1}{2}$ (b) $\frac{1}{3}$ (c) $\frac{2}{5}$ (d) $\frac{2}{7}$

(31) If the plane P_1 intercepted from positive parts of cartesian coordinates $2, 3, 5$ units and the plane P_2 intercepted from positive parts of cartesian coordinates $4, 6, 10$ units , then the two planes are .....

(a) coincident. (b) parallel.

(c) perpendicular. (d) intersecting and not perpendicular.

(32) Which of the following pairs of planes , their intersection line does parallel X-axis .....

(a) $X + 2y - z = 5$, $2X - y + 3z = 3$ (b) $X - 4y = 1$, $X - y = 6$

(c) $y - 2z = 3$, $2y + z = -1$ (d) $X - 3z = 4$, $2X + 7 = 0$

(33) Measure of the included angle between the plane $3X - 4y + 5z = 6$ and the plane Xy is .....

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{3}$ (c) $\frac{\pi}{4}$ (d) $\frac{\pi}{6}$

(34) If measure of the angle between the two planes $(3 , -4 , 2).\vec{r} = 7$, $3X + 4y - mz = 12$ is $90°$, then $m =$ .....

(a) $\frac{-7}{2}$ (b) $\frac{-3}{2}$ (c) $\frac{-25}{2}$ (d) $\frac{3}{2}$

(35) ***(1st Session 2023)*** If θ is the measure of the angle between the two planes :

$\vec{r} . (2 , -2 , 4) = 5$, $2X + y - 2z = 3$, then $\sin\theta =$ .....

(a) $-\frac{\sqrt{6}}{6}$ (b) $\frac{\sqrt{6}}{6}$ (c) $\frac{\sqrt{30}}{6}$ (d) $-\frac{\sqrt{30}}{6}$

(36) ***(1st Session 2021)*** If the measure of the angle between the two planes :

$X + y - 1 = 0$, $ky + z - 1 = 0$ equals $60°$, then $k =$ ..... , where $k > 0$

(a) 4 (b) $\frac{1}{2}$ (c) 2 (d) 1

(37) If the plane $2X - y + 2z = 4$ moves parallel to itself 3 units in $\overrightarrow{oX}$ direction , then its equation will be .....

(a) $2X - y + 2z = 1$ (b) $2X - y + 2z = 7$

(c) $2X - y + 2z = 10$ (d) $2X - y + 2z = -7$

(38) If the plane $\vec{r} . (\frac{1}{2} , \frac{-1}{2} , \frac{1}{\sqrt{2}}) = 3$ is translated paralleling itself away from the origin point , then its equation may become .....

(a) $\vec{r} . (1 , -1 , \sqrt{2}) = 2$ (b) $\vec{r} . (3 , -3 , \sqrt{6}) = 12$

(c) $\vec{r} . (1 , -1 , \sqrt{2}) = 6$ (d) $\vec{r} . (1 , -1 , \sqrt{2}) = 8$

(39) The distance between the z-axis and the plane $\frac{X}{a} + \frac{y}{b} = 1$ is ..... length unit. where $ab > 0$

(a) ab (b) $\sqrt{a^2 + b^2}$ (c) $\frac{ab}{\sqrt{a^2 + b^2}}$ (d) $\frac{a + b}{\sqrt{a^2 + b^2}}$

(40) *(1st Session 2024)* The length of perpendicular drawn from the point $(0 , 12 , 0)$ to the plane which passes through the point $(1 , 0 , 2)$ and perpendicular to the straight line whose equation $\vec{r} = (2 , 5 , 6) + t(-3 , -4 , 0)$ equals ..... length unit.

(a) 5 (b) $\frac{33}{5}$ (c) 9 (d) $\frac{51}{5}$

7 **Find the equation of the line of intersection of each pair of the following planes :**

(1) $3x - y + 2z = 3$, $x - 2y + 5z = 2$

(2) $x + 2y - 2z = 1$, $2x + y - 3z = 5$

(3) $\vec{r} \cdot (3\hat{i} - \hat{j} + \hat{k}) = 1$, $\vec{r} \cdot (\hat{i} + 4\hat{j} - 2\hat{k}) = 2$

8 Find the different forms of the equation of the intersection line of the two planes $x + y + z = 1$ and $x + z = 0$

9 Prove that the two planes $3x - y - z = 2$, $x - 2y + 2z = 1$ are intersecting and find the equation of the intersecting line.

10 Prove that the intersection line of the two planes $x + y - z = 0$, $x - y - 5z + 7 = 0$ is parallel to the straight line whose cartesian equation $\frac{x+3}{3} = \frac{y-1}{-2} = \frac{z-5}{1}$

11 Find the equation of the plane which is perpendicular to each of the two planes $x + 2y - 3z + 5 = \text{zero}$, $3x - y + 2z - 1 = \text{zero}$ and passes through the point $(1 , 3 , -5)$

12 Find the equation of the plane passes through the two points $(1 , -1 , 1)$, $(-1 , 1 , 1)$ and perpendicular to the plane $x + 2y + 2z = 5$

13 Find the equation of the plane passing through the intersection line of the two planes $x + y + 2z + 1 = 0$, $2x + y - z + 1 = 0$ and is parallel to the straight line passing through the two points $m_1 (2 , 5 , -3)$ and $m_2 (3 , -2 , 2)$

14 Find the equation of the plane passing through the intersection line of the two planes $x + 2y + 3z - 4 = 0$, $2x + y - z + 8 = 0$ and is perpendicular to the plane $3x + 2y + z + 6 = 0$

15 Find the equation of the plane passing through the points $(2\ ,-3\ ,2)$ and contains the intersection line of the two planes $6x+4y+3z+5=0$ and $2x+y+z-2=0$

16 Find the distance between the point $(3\ ,-5\ ,2)$ and the plane $8x-2y+z=5$

« $\frac{31}{\sqrt{69}}$ length unit »

17 Find the length of the perpendicular drawn from the point $(1\ ,-1\ ,3)$ to the plane whose equation is $\vec{r}\ .\ (2\ ,2\ ,-1)=5$ « $\frac{8}{3}$ length unit »

18 Find the distance between point $(2\ ,1\ ,-1)$ and the plane $\vec{r}\ .\ (\hat{i}-2\hat{j}+4\hat{k})=9$

« $\frac{13}{\sqrt{21}}$ length unit »

19 If the length of the perpendicular drawn from the point A $(0\ ,-1\ ,2)$ to the plane $\sqrt{2}x+y-z+k=0$ equals 2 unit length , find the value of k « 7 or – 1 »

20 Find the point which lies on the x-axis and equidistant from each of the two planes $12x-16y+15z+1=0$, $2x+2y-z-1=0$ « $(2\ ,0\ ,0)$ or $(\frac{11}{43}\ ,0\ ,0)$ »

21 *(2nd Session 2017)* Find the equation of the plane that parallels the plane $2x+y-4z=0$ and at a distance $\sqrt{21}$ length unit from the point $(1\ ,2\ ,0)$

22 Calculate the distance between the point $(-1\ ,1\ ,-2)$ and the plane passing through the three points $M_1\ (1\ ,-1\ ,1)$, $M_2\ (-2\ ,1\ ,3)$, $M_3\ (4\ ,-5\ ,-2)$ « 4 length unit »

23 Find the distance between the two parallel planes $x-2y+2z=1$ and $2x-4y+4z=3$ « $\frac{1}{6}$ length unit »

24 Prove that the two planes $2x+y+2z=8$, $4x+2y+4z+5=0$ are parallel , then find the distance between them. « $3\frac{1}{2}$ length unit »

25 Prove that the distance between the two parallel planes $P_1: ax+by+cz+d_1=0$, $P_2=ax+by+cz+d_2=0$ is $\frac{|d_2-d_1|}{\sqrt{a^2+b^2+c^2}}$

26 Find the distance between the straight line $\vec{r}=(2\ ,-2\ ,3)+t\ (1\ ,-1\ ,4)$ and the plane $\vec{r}\ .\ (1\ ,5\ ,1)=5$ « $\frac{10}{3\sqrt{3}}$ length unit »

27 **If the points A , B , C , D are in space whose position vectors are $-\hat{j}+\hat{k}$, $2\hat{i}-\hat{j}+3\hat{k}$, $-\hat{i}-2\hat{j}+2\hat{k}$, $7\hat{i}-4\hat{j}+2\hat{k}$ respectively :**

(1) Find the normal vector to the plane ABC

(2) Show that the length of the perpendicular from D to the plane ABC equals $2\sqrt{6}$

(3) Show that the two planes ABC , DBC are orthogonal.

(4) Find the equation of the line of the intersection of the two planes ABC , ODB

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28 **If the plane X contains points A (1 , 4 , 2) , B (1 , 0 , 5) , C (0 , 8 , – 1) and the plane Y contains point D (2 , 2 , 3) and the vector $\vec{n}=\hat{i}+2\hat{j}+2\hat{k}$ is perpendicular to it :**

(1) Find the cartesian equation of X

(2) Find the cartesian equation of the plane Y

(3) What is the values of t , f if point (t , 0 , f) belongs to each of the two planes X , Y ?

(4) Find the vector equation of the line of intersection of the two planes X , Y

(5) If point (1 , 1 , p) equidistant from the two planes X , Y , find all possible values of p

Higher skills

29 Find the equation of the plane which bisects the angle between the two planes

$7x+y-6=0$, $3x+5y-4z+1=0$